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Frame assembly and button

Abstract

A frame assembly is adapted to be mounted on a substrate having a contact switch. The frame assembly includes a frame body and at least one spring arm. The frame body is adapted to move along an operational direction to be in contact with the contact switch. The spring arms are connected to the frame body and obliquely extend from the frame body to the substrate, so that when the frame body is not pressed by an external force, there is a spacing between the frame body and the contact switch. In this way, the number of parts of a button may be reduced by connecting the spring arms to the frame body. The present invention further provides a button using the above frame assembly.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10304641	12/2018	Watanabe	N/A	H01H 13/14
10930448	12/2020	Ito	N/A	H01H 13/14
2024/0347289	12/2023	Chen	N/A	H01H 13/023

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101419877	12/2008	CN	N/A
101447357	12/2008	CN	N/A
208368389	12/2018	CN	N/A
109509667	12/2019	CN	N/A
214670983	12/2020	CN	N/A
H10255587	12/1997	JP	N/A
2001184978	12/2000	JP	N/A
2005353409	12/2004	JP	N/A
M298179	12/2005	TW	N/A
I402880	12/2012	TW	N/A

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to a frame assembly and a button, and more particularly to a frame assembly applied to a button capable of emitting light, and a button using the frame assembly.

BACKGROUND OF THE INVENTION

(2) A button is a very common input device at present, which allows a user to input a signal by press to control large machinery such as an elevator. To allow the user to conveniently identify the position or type of the button, at present, light-emitting elements will be mounted inside some buttons, for example, on substrates at the bottoms of the buttons, and light-transmitting windows are arranged on the buttons to facilitate the user's observation.

(3) The known button includes a frame and an electrical signal generation component (such as a pressure sensor and a pressure switch). In order not to trigger the electrical signal generation component when the user does not press the frame, and to make the frame generate a rebound

function after the press is over, or to give the user a press forced feedback during the press and other reasons, a reset component such as a spring which can separate the frame from the electrical signal generation component when the frame is pressed and may rebound after the press is over must be arranged in the button. To prevent the displacement or loss of the reset component during assembly or maintenance, additional components must be designed to fix it, thus complicating the structure of the button and increasing the troubles of production and maintenance.

SUMMARY OF THE INVENTION

(4) The present invention provides a frame assembly, to reduce the number of components required to constitute a button.

(5) In addition, the present invention provides a button, to reduce the number of components.

(6) To achieve the above advantages, an embodiment of the present invention provides a frame assembly adapted to be mounted on a substrate having a contact switch. The frame assembly includes a frame body and at least one spring arm. The frame body is adapted to move along an operational direction to be in contact with the contact switch. The at least one spring arm is connected to the frame body. The at least one spring arm obliquely extends from the frame body to the substrate, so that when the frame body is not pressed by an external force, there is a spacing between the frame body and the contact switch.

(7) In an embodiment of the present invention, the above frame body has a ring wall forming an opening, and the at least one spring arm is connected to a ring side wall surface of the ring wall.

(8) In an embodiment of the present invention, the above frame body has a ring wall forming an opening, and the at least one spring arm is located inside or outside the opening.

(9) In an embodiment of the present invention, each of the above at least one spring arm includes a first extension section and a second extension section, the first extension section is connected between the second extension section and the frame body, and the first extension section and the second extension section have different extension directions.

(10) In an embodiment of the present invention, the above frame assembly further includes an outer frame being adapted to be mounted on the substrate and surrounding the frame body, the outer frame has a slide groove, an extension direction of the slide groove is parallel to the operational direction, and the frame body is provided with a bump slidably arranged in the slide groove.

(11) The above button provided by the present invention includes a substrate and a frame assembly. The substrate has a contact switch. The frame assembly is connected to the substrate. The frame assembly includes a frame body and at least one spring arm. The frame body is adapted to move along an operational direction to be in contact with the contact switch. The spring arms are connected to the frame body and obliquely extend from the frame body to the substrate, so that when the frame body is not pressed by an external force, there is a spacing between the frame body and the contact switch.

(12) In an embodiment of the present invention, the above frame body has a ring wall forming an opening, and the at least one spring arm is connected to a ring side wall surface of the ring wall.

(13) In an embodiment of the present invention, the above frame body has a ring wall forming an opening, and the at least one spring arm is located inside or outside the opening.

(14) In an embodiment of the present invention, each of the above spring arms includes a first extension section and a second extension section, the first extension section is connected between the second extension section and the frame body, and the first extension section and the second extension section have different extension directions.

(15) In an embodiment of the present invention, the above button further includes an outer frame being adapted to be mounted on the substrate and surrounding the frame body, the outer frame has a slide groove, an extension direction of the slide groove is parallel to the operational direction, and the frame body is provided with a bump slidably arranged in the slide groove.

(16) In an embodiment of the present invention, the above substrate further includes at least one light-emitting element, and a position of the at least one spring arm projected on the substrate along

the operational direction avoids a position of the at least one light-emitting element.

(17) Based on the above description, in the frame assembly and the button using the same according to the present invention, the frame body is supported by the spring arms connected to the frame body, so that when the frame body does not bear an external force, there is a spacing between the frame body and the contact switch on the substrate. When the frame body bears an external force and moves towards the contact switch along the operational direction, the contact switch can be triggered by deforming the spring arms, and after the frame body does not bear an external force, the above spacing is generated again with an action of returning the spring arms to the original shape, thus reducing the number of components required for the button.

(18) To make the above and other objectives, features and advantages of the present invention more obvious and understandable, the following is a detailed description of the embodiments in combination with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will become more readily apparent to those ordinarily skilled in the art after reviewing the following detailed description and accompanying drawings, in which:

(2) FIG. 1 is a schematic diagram of an exploded view of a button according to an embodiment of the present invention;

(3) FIG. 2A is a schematic diagram of a side view of a frame body in FIG. 1;

(4) FIG. 2B is a schematic diagram of a top view of the frame body in FIG. 1;

(5) FIG. 3 is a schematic diagram of a partial sectional view of the button in FIG. 1;

(6) FIG. 4 is a schematic diagram of a partial sectional view of the button in FIG. 1 during actuation;

(7) FIG. 5 is a schematic diagram of a frame according to another embodiment of the present invention; and

(8) FIG. 6 is a schematic diagram of a frame according to another embodiment of the present invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(9) The present invention will now be described more specifically with reference to the following embodiments. It is to be noted that the following descriptions of preferred embodiments of this invention are presented herein for purpose of illustration and description only. It is not intended to be exhaustive or to be limited to the precise form disclosed.

(10) Hereinafter, the terms used in the description of the embodiments according to the present invention, such as the description of orientational or positional relationships indicated by “upper”, “lower”, and the like, are described according to the orientational or positional relationships shown in the figures used. The above terms are merely for the convenience of describing rather than limiting the present invention, that is to say, they do not indicate or imply that the element mentioned must have a specific orientation or be constructed in a specific orientation. In addition, the terms “first”, “second”, and the like mentioned in this specification or the scope of the patent application are only used to name the elements or distinguish different embodiments or scopes, rather than to limit the upper or lower limit of the number of elements.

(11) FIG. 1 is a schematic exploded view of a button according to an embodiment of the present invention. FIG. 2A is a schematic side view of a frame body in FIG. 1. FIG. 2B is a schematic top view of the frame body in FIG. 1. FIG. 3 is a schematic partial sectional view of the button in the embodiment of FIG. 1. FIG. 4 is a schematic partial sectional view of the button in the embodiment of FIG. 1 during actuation. Referring to FIG. 1 to FIG. 4, a button 1 in this embodiment includes a substrate 2 and a frame assembly 3. The substrate 2 has a contact switch 21. The frame assembly 3

is connected to the substrate **2** and includes a frame body **31** and spring arms **32**. The frame body **31** is adapted to move along an operational direction **D** to be in contact with the contact switch **21**. The spring arms **32** are connected to the frame body **31** and obliquely extend from the frame body **31** to the substrate **2**, so that when the frame body **31** is not pressed by an external force, there is a spacing **S** between the frame body **31** and the contact switch **21**.

(12) When the frame body **31** bears the external force and is extruded towards the contact switch **21** along the operational direction **D**, the spring arms **32** will deform during the process, meanwhile, the frame body **31** moves towards the substrate **2** to reduce the spacing **S** between the frame body **31** and the contact switch **21**, and finally the contact switch **21** is pressed (as shown in FIG. 4). When the external force disappears, the spring arms **32** return to the original shape; and by pushing against the substrate **2**, there is the spacing **S** between the frame body **31** and the contact switch **21** again (as shown in FIG. 3).

(13) As shown in FIG. 1, in this embodiment, for example, a plurality of light-emitting elements **22** are distributed on the substrate **2**. The light-emitting elements **22** are, for example, light-emitting diodes (LEDs), but are not limited to this. The contact switch **21** is, for example, a push switch mounted on the substrate **2**. The number of contact switches may be, for example, two, but is not limited to this. In this embodiment, the button **1**, for example, is a button of an elevator. A user presses the frame body **31** indirectly through a transparent frame (not shown in figure) along the operational direction **D** to make the frame body **31** contact with the contact switch **21**, so that a circuit where the contact switch **21** is located may be connected to generate an electrical signal to control the operation of the elevator.

(14) As shown in FIG. 1, the frame assembly **3**, for example, further includes an outer frame **33** and a front cover **34**. The front cover **34** is mounted on the outer frame **33**, and the outer frame **33** is mounted on the substrate **2** and surrounds the frame body **31** and the spring arms **32**. The outer frame **33** cooperates with the front cover **34** to form a first opening **331** which is open along the operational direction **D**. The first opening **331** has an inner wall surface **332**, and for example, a slide groove **333** is provided on the inner wall surface **332** of the outer frame **33**. An extension direction of the slide groove **333** is parallel to the operational direction **D**.

(15) The frame body **31** has, for example, a ring wall **311**. The ring wall **311** defines a second opening **312** which is opened along the operational direction **D**. Press portions **313** adapted to press two contact switches **21** along the operational direction **D** are arranged at positions of the ring wall **311** that correspond to the two contact switches **21**. Referring to FIG. 2A and FIG. 2B, the shape of the frame body **31** is not specially restricted, may be a square shown in FIG. 2B, a triangle, or a circle, and may be determined, for example, by the shape of the button **1** or the positions of the light-emitting elements **22** on the substrate **2**.

(16) In addition, referring to FIG. 1 to FIG. 2B, a bump **314** corresponding to the slide groove **333** arranged on the above outer frame **33** in position is provided on a side wall surface **311a** of one side of the frame body **31** that is away from the second opening **312**, for example. When the frame body **31** is combined with the outer frame **33**, the bump **314** enters the slide groove **333**, so that the frame body **31** is restricted by the outer frame **33** and can only slide along the operational direction **D**.

(17) As shown in FIG. 1 to FIG. 4, for example, the spring arms **32** are connected to the side wall surface **311a** of the ring wall **311** that is outside the second opening **312**; for example, as shown in FIG. 2A and FIG. 2B, the spring arms are connected to the side wall surface **311a** and close to the bottom of the side wall surface **311a**; and for example, the spring arms extend to the substrate **2** from four corners of the frame body **31**, but are not limited to this. In other words, in other embodiments, the spring arms **32** may extend to the substrate **2** at the top of the side wall surface **311a**, or may extend to the substrate at a position between the top and bottom of the side wall surface **311a**, or may obliquely extend to the substrate **2** with a bottom surface **315** (see FIG. 2A) of the frame body **31** that faces the substrate **2** as a joint between the spring arms **32** and the frame

body **31**. Through the above structure, when the frame body **31** and the substrate **2** are combined, the spacing **S** can be formed between the frame body **31** and the contact switch **21**, as shown in FIG. **3**.

(18) In addition, the positions of the spring arms **32** may be determined, for example, by the distribution of the light-emitting elements **22** on the substrate **2**, that is to say, the position of the spring arms **32** projected on the substrate **2** along the operational direction **D** will avoid the position of the at least one light-emitting element **22**. Therefore, in the embodiment shown in FIG. **1** where the plurality of light-emitting elements **22** on the substrate **2** may be seen in the second opening **312**, the spring arms **32** are located outside the second opening **312**, and the plurality of light-emitting elements **22** may be seen between the outer frame **33** and the frame body **31**, but in the embodiment where there are fewer light-emitting elements **22** in the second opening **312** of the frame body **31**, for example, as shown in FIG. **5**, the spring arms **32** may be arranged on a side wall surface **311b** in the second opening **312**. Similarly, there is no restriction that the spring arms **32** must be arranged at the corners of the frame body **31** as in FIG. **1** to FIG. **5**, and the spring arms may be arranged on the edges of the frame body **31**. The number of spring arms **32** is not specially limited.

(19) Because the spring arms **32** deform when the frame body **31** bears the external force acting along the operational direction **D**, and return to the original shape when the frame body **31** does not bear the external force, the spring arms **32** may be, for example, made of an elastic material. In the embodiment where the frame body **31** is made of an elastic material, both the spring arms **32** and the frame body **31** may be integrally formed.

(20) FIG. **6** is a schematic diagram of a frame according to another embodiment of the present invention. Referring to FIG. **6**, in the embodiment shown in FIG. **6**, spring arms **32a** are different from the spring arms **32** in FIG. **1**, FIG. **2A**, and FIG. **3**. Specifically, the spring arms **32** in FIG. **1**, FIG. **2A**, and FIG. **3** obliquely extend to the substrate **2** linearly from the side wall surface **311a** of the frame body **31** directly. Each of the spring arms **32a** in FIG. **6**, for example, includes a first extension section **321** and a second extension section **322**. The first extension section **321** is connected between the second extension section **322** and the frame body **31**, and extends out along a direction parallel to the substrate **2**, for example. The second extension section **322** obliquely extends to the substrate **2**, for example. In other words, the first extension section **321** and the second extension section **322** may extend in different directions, but a contact point of the spring arm **32a** composed of the first extension section **321** and the second extension section **322**, that is connected to the frame body **31**, and a contact point of the spring arm **32a** that is in contact with the substrate **2** are still inclined to the operational direction **D**.

(21) Based on the above description, in the frame assembly and the button using the same according to the present invention, the frame body is supported by the spring arms connected to the frame body, so that when the frame body does not bear an external force, there is the spacing between the frame body and the contact switch on the substrate. When the frame body bears an external force and moves towards the contact switch along the operational direction, the contact switch can be triggered by deforming the spring arms, and after the frame body does not bear the external force, the above spacing is generated again with an action of returning the spring arms to the original shape. The number of components required for the button is reduced by combining the frame body for pressing the contact switch with the spring arms for driving the frame body to rebound, and the spring arms are capable of flexibly making changes corresponding to the positions of the light-emitting elements on the substrate.

(22) While the invention has been described in terms of what is presently considered to be the most practical and preferred embodiments, it is to be understood that the invention needs not be limited to the disclosed embodiment. On the contrary, it is intended to cover various modifications and similar arrangements included within the spirit and scope of the appended claims which are to be

accorded with the broadest interpretation so as to encompass all such modifications and similar structures.

Claims

1. A frame assembly adapted to be mounted on a substrate having a contact switch, wherein the frame assembly comprises: a frame body adapted to move along an operational direction to be in contact with the contact switch; and at least one spring arm connected to the frame body, wherein the at least one spring arm obliquely extends from the frame body to the substrate, so that when the frame body is not pressed by an external force, there is a spacing between the frame body and the contact switch; wherein the frame body has a ring wall forming an opening, and the at least one spring arm is connected to a ring side wall surface of the ring wall.
 2. The frame assembly according to claim 1, wherein the at least one spring arm is located inside or outside the opening.
 3. The frame assembly according to claim 1, wherein each of the at least one spring arm comprises a first extension section and a second extension section, the first extension section is connected between the second extension section and the frame body, and the first extension section and the second extension section have different extension directions.
 4. The frame assembly according to claim 1, further comprising an outer frame being adapted to be mounted on the substrate and surrounding the frame body, wherein the outer frame has a slide groove, an extension direction of the slide groove is parallel to the operational direction, and the frame body is provided with a bump slidably arranged in the slide groove.
 5. A button, comprising: a substrate having a contact switch; a frame assembly connected to the substrate, wherein the frame assembly comprises: a frame body adapted to move along an operational direction to be in contact with the contact switch; and at least one spring arm connected to the frame body, wherein the at least one spring arm obliquely extends from the frame body to the substrate, so that when the frame body is not pressed by an external force, there is a spacing between the frame body and the contact switch; wherein the frame body has a ring wall forming an opening, and the at least one spring arm is connected to a ring side wall surface of the ring wall.
 6. The button according to claim 5, wherein the at least one spring arm is located inside or outside the opening.
 7. The button according to claim 5, wherein each of the at least one spring arm comprises a first extension section and a second extension section, the first extension section is connected between the second extension section and the frame body, and the first extension section and the second extension section have different extension directions.
 8. The button according to claim 5, further comprising an outer frame being adapted to be mounted on the substrate and surrounding the frame body, wherein the outer frame has a slide groove, an extension direction of the slide groove is parallel to the operational direction, and the frame body is provided with a bump slidably arranged in the slide groove.
 9. The button according to claim 5, wherein the substrate further comprises at least one light-emitting element, and a position of the at least one spring arm projected on the substrate along the operational direction avoids a position of the at least one light-emitting element.
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